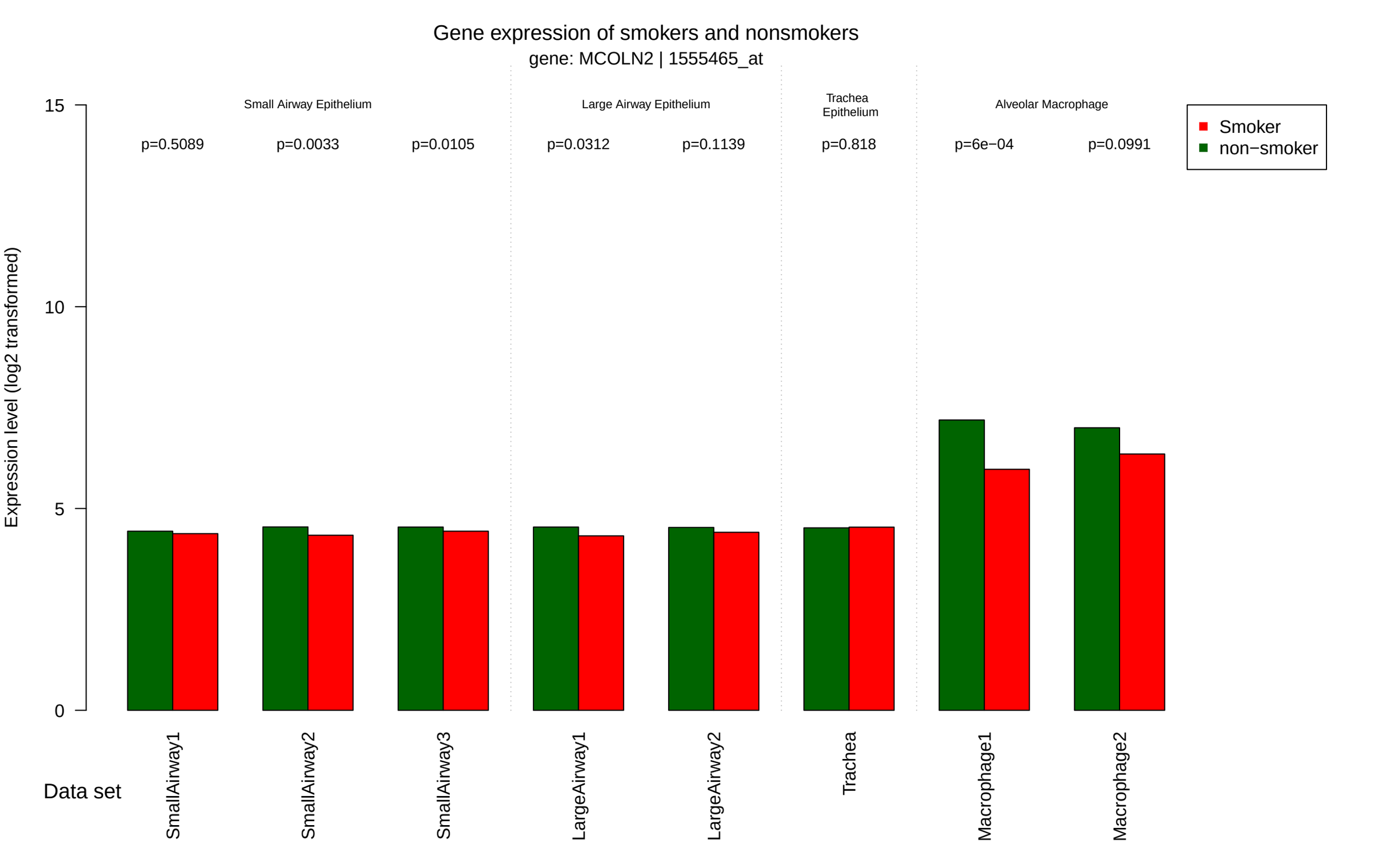
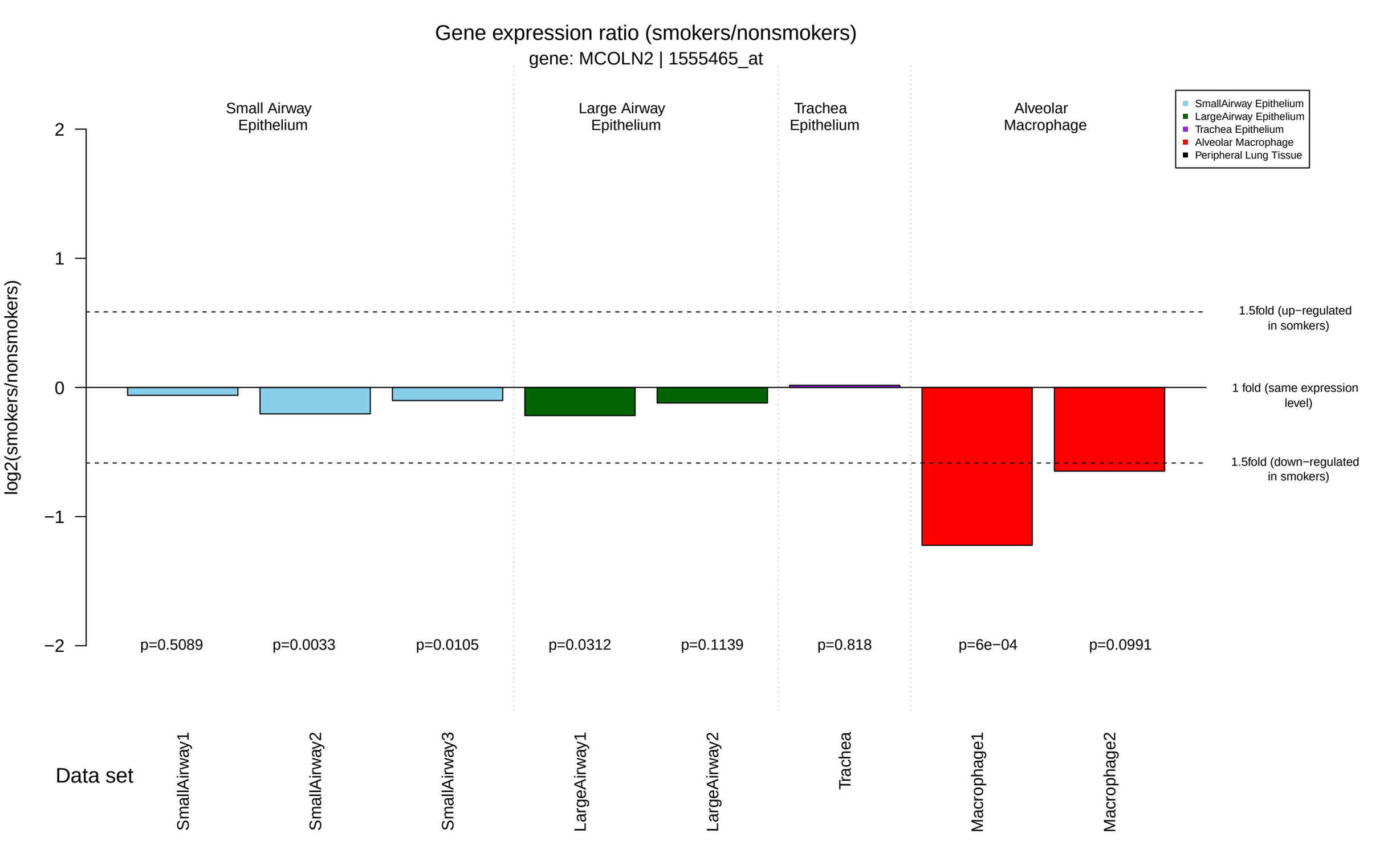
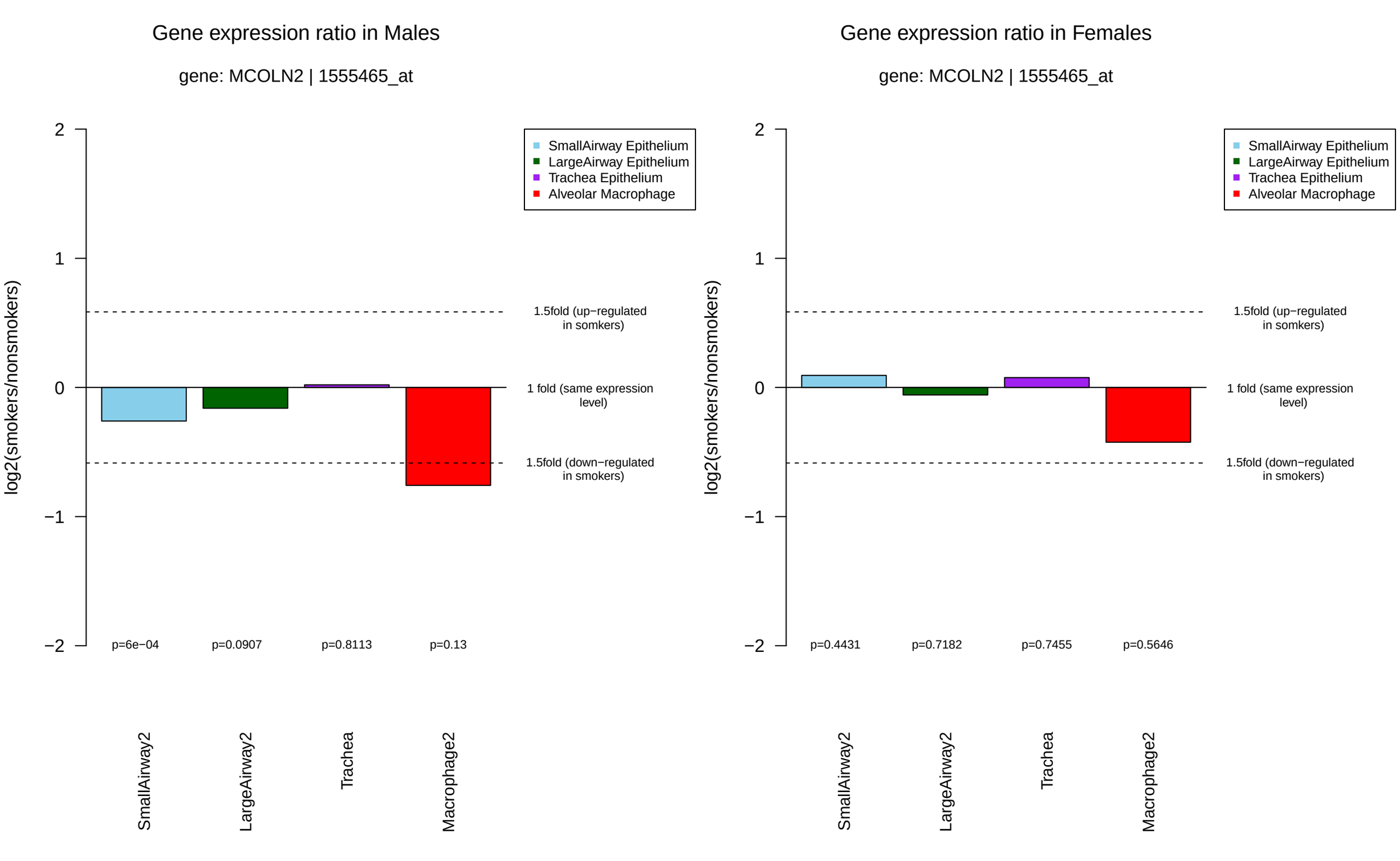
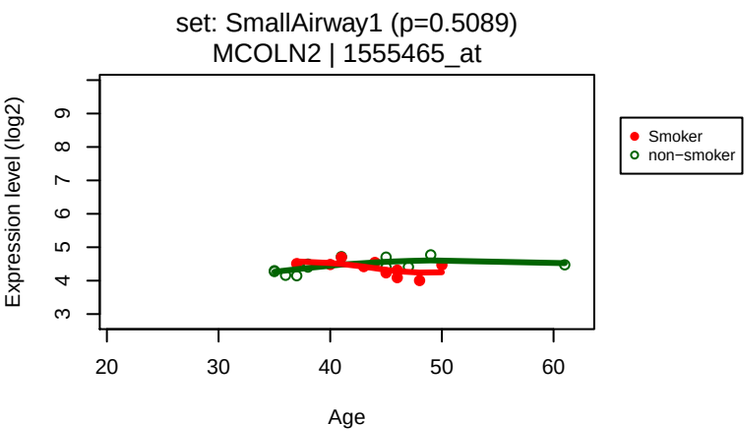
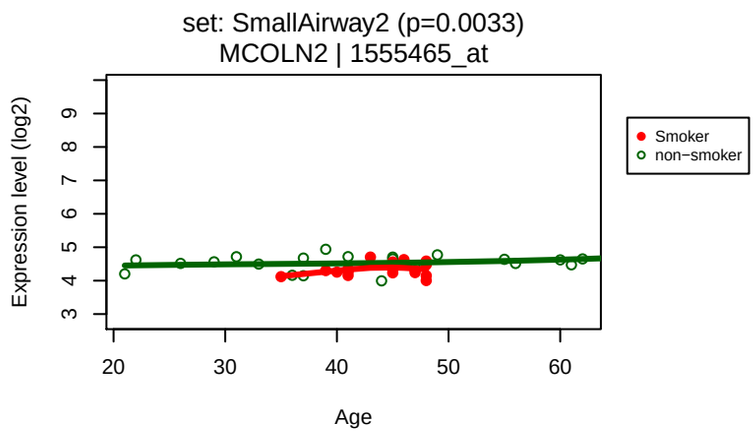
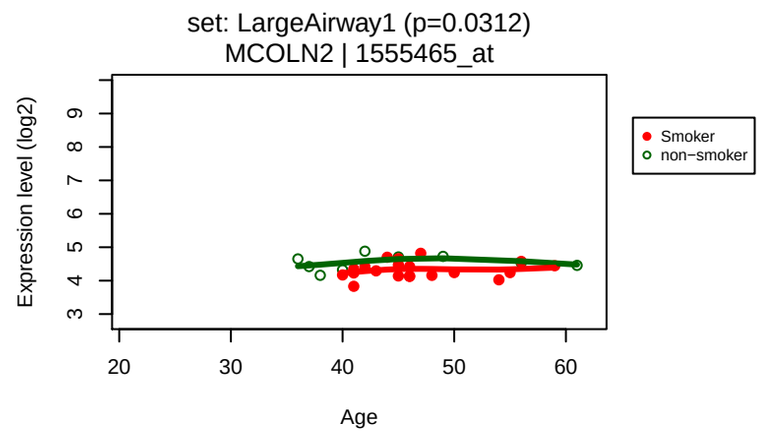

Gene expression of smokers and nonsmokers
gene: MCOLN2 | 1555465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5089
p=0.0033
p=0.0105
p=0.0312
p=0.1139
p=0.818
p=6e−04
p=0.0991
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

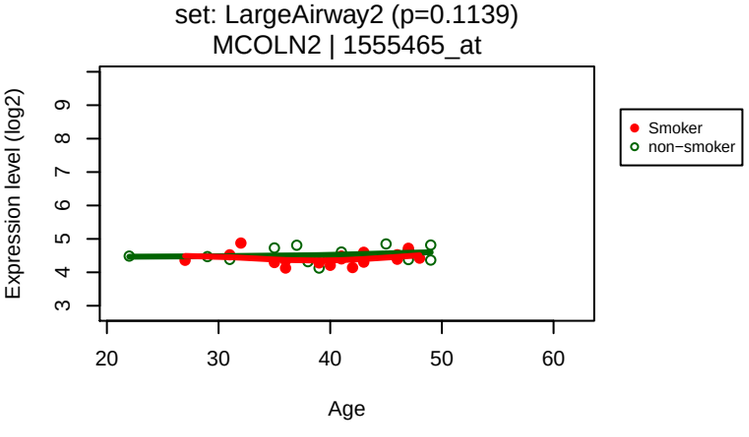

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN2 | 1555465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5089
p=0.0033
p=0.0105
p=0.0312
p=0.1139
p=0.818
p=6e−04
p=0.0991
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

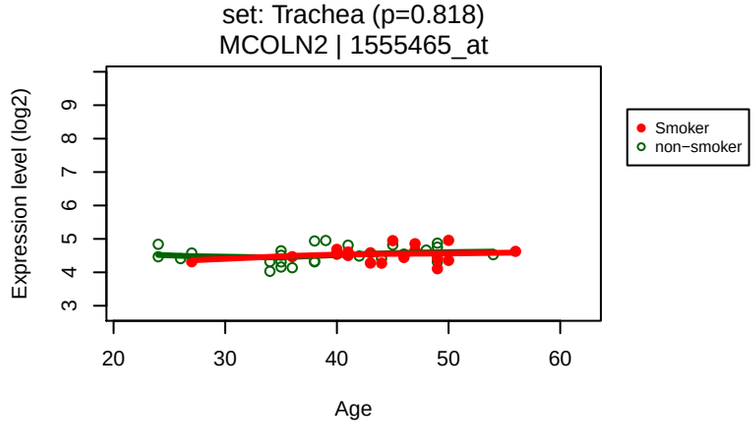

Gene expression ratio in Males
gene: MCOLN2 | 1555465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.0907
p=0.8113
p=0.13
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN2 | 1555465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4431
p=0.7182
p=0.7455
p=0.5646
SmallAirway2
LargeAirway2
Trachea
Macrophage2

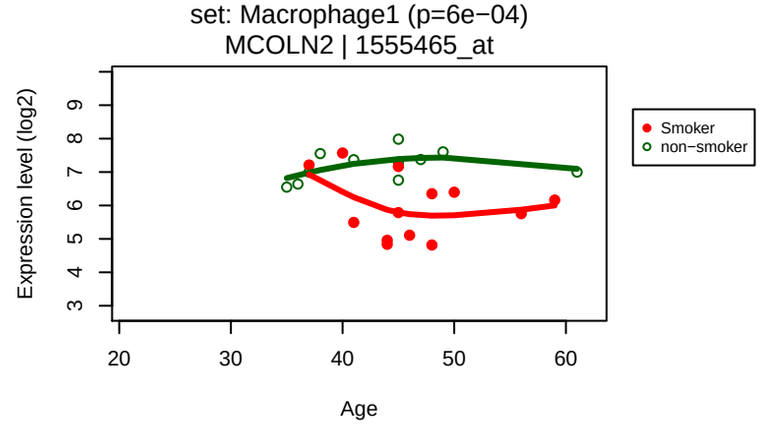

set: SmallAirway1 (p=0.5089)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker

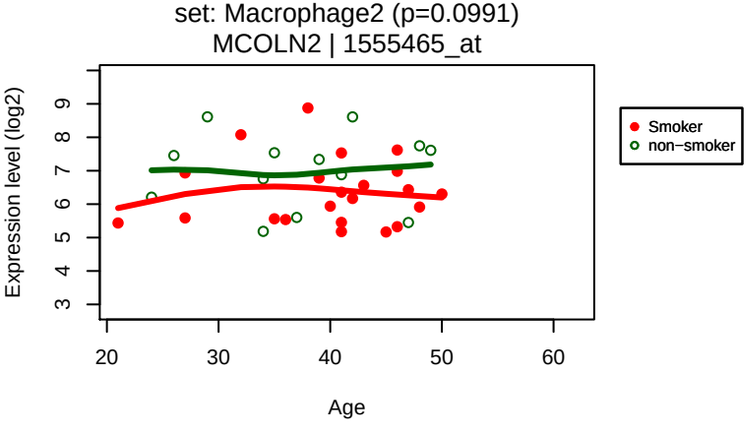

set: SmallAirway2 (p=0.0033)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker

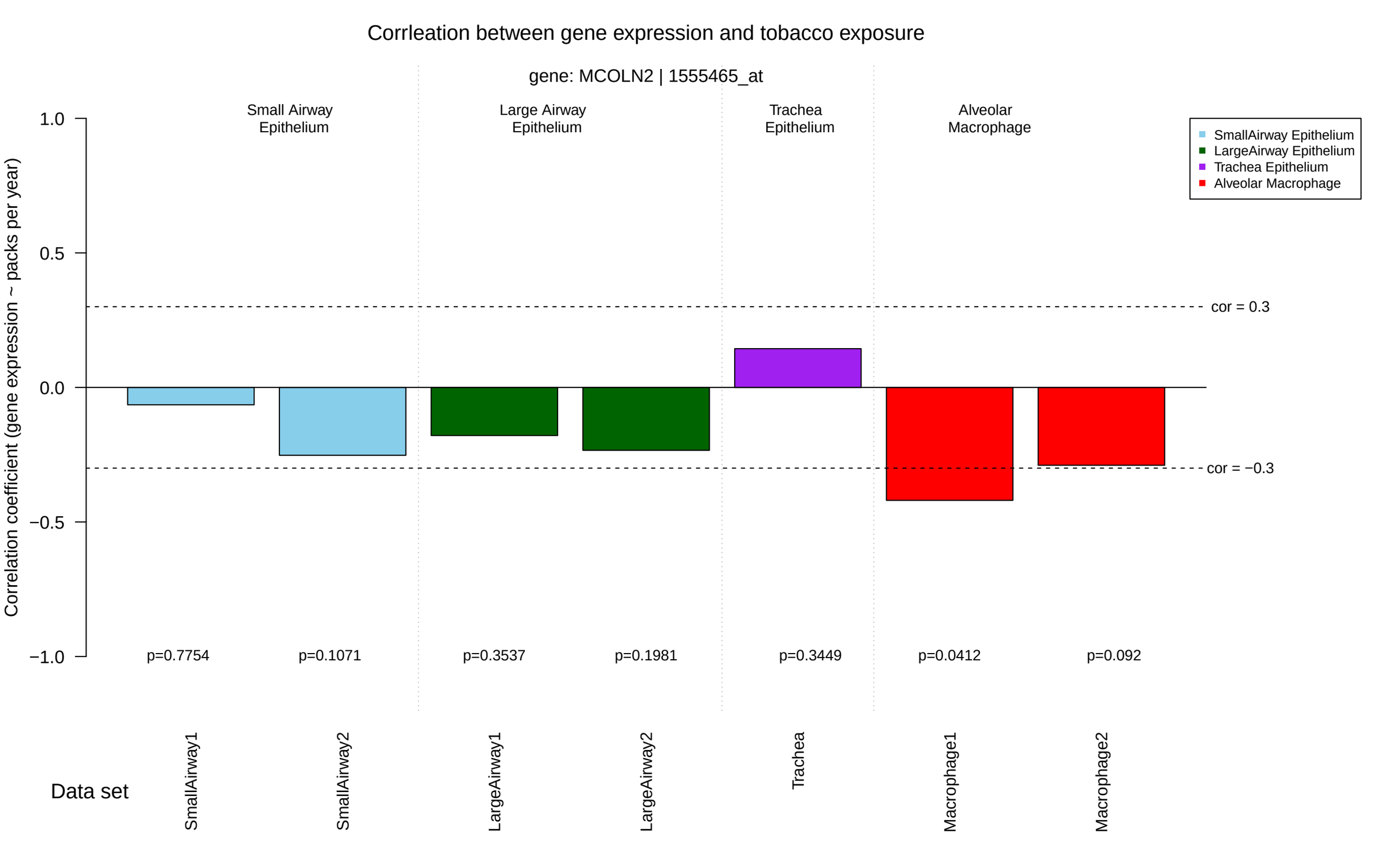

set: LargeAirway1 (p=0.0312)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1139)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.818)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=6e−04)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0991)
MCOLN2 | 1555465_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN2 | 1555465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7754
p=0.1071
p=0.3537
p=0.1981
p=0.3449
p=0.0412
p=0.092
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2